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10.5.5 Nesting of loops

A loop which contains another loop is called as a nested loop.

The syntax is given below:

```
for (initialization(s); test-expression; update expression(s))
{
        for (initialization(s); test-expression; update expression(s)
        {
        statement(s);
         }
statement(s);
}
```

```
while(condition)
{
        while(condition)
        {
        statement(s);
        }
statement(s);
}
```

```
do
{
statement(s);
        do
        {
        statement(s);
        }while(condition);
} while( condition );
```

Illustration 10.17 C++ program to display matrix multiplication table using nested for loop

```
#include<iostream>
using namespace std;
int main(void)
{
    cout<< "A multiplication table:" <<endl <<"  1\t2\t3\t4\t5\t6\t7\t8\t9" <<endl<< "" <<endl;
    for(int c = 1; c < 10; c++)
    {
    cout<< c << "| ";
    for(int i = 1; i< 10; i++)
    {
    cout<<i * c << '\t';
    }
    cout<<endl;
    }
return 0;
}
```